Vaishant Makan

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Education

University College London (UCL) Sept 2026 – Sept 2027
MSc Artificial Intelligence and Data Engineering *London, UK*
- Recipient of UCL India Excellence Scholarship

Thapar Institute of Engineering and Technology June 2019 – June 2023
Bachelor of Engineering in Computer Science and Engineering *Patiala, India*
- **Specialization:** Conversational AI (NVIDIA Collaboration)
- Academic Merit Scholarship Awardee — **GPA: 8.76/10.0**
- Selected coursework: Algorithms, Data Structures, Operating Systems, Databases, Machine Learning

Work Experience

ION Group Jan 2023 – May 2026
Software Developer – Fixed Income Connectivity *Noida, India*
- Design and develop **low-latency C++ market connectivity gateways** for US, European, and APAC exchanges and financial institutions, implementing **FIX 4.x/5.x**, proprietary binary protocols **(BOP/BIMP for FMX UST)**, IBM WebSphere MQ, FpML, and SWIFT; supporting **20M+ daily transactions**.
- Build **server-side infrastructure** for **front-office fixed-income trading** across Market Data, Order/Quote/Trade Management, Reference Data, and STP, achieving **sub-millisecond latency** through **multithreading and optimised protocol parsing** in Dealer-to-Dealer (D2D) markets.
- **Led XTP-AI Initiative:** Developed LLM-powered automated unit test generation system for ION's 100K+ LOC modular trading codebase; architected Jenkins-integrated pipeline performing AST parsing (Tree-sitter), dependency analysis, and LLM orchestration to produce context-aware tests (Mockito/TestNG); implemented fail-fast retry mechanisms and self-improving feedback loops, **reducing manual test authoring effort by 50%** while maintaining **80%+ test coverage**.
- Designed and built an LLM-driven workflow **(LangChain + Qwen3)** that processes support tickets, auto-classifies them, recommends resolutions, and performs post-resolution sentiment analysis.
- Implemented **CI/CD pipelines** (Jenkins, Git, Docker, Ansible) to automate builds and deployments; cut release failures by **60%** and accelerated time-to-market by **70%**.

Krash IT Services Sept 2021 – Dec 2021
iOS Developer Intern *Remote*
- Developed iOS applications in **Swift, SwiftUI, Objective-C** using **Cocoa Touch** and **Realm**, delivering features across iPhone, iPad, and Apple Watch.
- Collaborated on UI/UX with **Figma** and improved app performance by fixing bottlenecks and bugs.

Projects

Face Recognition Attendance System | *GitHub* **Python, OpenCV, Tkinter, MySQL**
- Built a **multi-role desktop app** with Tkinter and MySQL authentication, enabling students, teachers, and admins to log in, manage profiles, and track attendance.
- Implemented **real-time face recognition** (OpenCV + LBPH) with automated dataset collection and MySQL integration to generate accurate attendance.

Real-Time Online Chess Application | *GitHub* **TypeScript, WebSockets, Prisma, PostgreSQL**
- Built a **full-stack, real-time chess platform** with a TypeScript-based **Turborepo** monorepo, featuring user authentication, persistent profiles, and live gameplay.
- Developed a responsive **React + Tailwind** frontend, a scalable **Node.js/Express** backend, and a dedicated **WebSocket** server for bidirectional game-state synchronization.

Technical Skills

Programming: C++, Python, TypeScript, JavaScript
Frameworks: PyTorch, Scikit-learn, Numpy, Pandas, React, Node.js, Express.js
Databases: PostgreSQL, MySQL
Tools & Platforms: Linux, Jenkins, Docker, Kubernetes, Ansible, Git